Motor Sequence Learning Experiment Report

This Report is prepared by Ashmita Ukil (AU2320042) for PSY310: Lab in Psychology

Name of the Experiment: Motor Sequence Learning

Course name: PSY310: Lab in Psychology

Date: 1st October 2025

Name of the student: Ashmita Ukil

Enrollment no.: AU2320042

GitHub Link: *https://github.com/ukilashmita-bot/Motor-sequence-Learning.git*

Description

The participants were conditioned in this experiment to match and predict the presence of particular patterns or series of visual stimuli. Their sequential information acquisition and retention was tested by comparing the reaction times (RTs) in a sequential and random stimulus condition. The experiment compares sequence learning as an eye into cognitive process of learning, prediction and memory consolidation.

Purpose

The study is meant to examine how human beings learn, remember and generalise sequential information. The research is informative about the cognitive mechanisms involved in skill learning and the consolidation of implicit memory as it investigates reaction time in sequence and random trials.

Introduction

Closely related to learning is contingency prediction between events or between stimuli and responses. In the case of a predictable contingency, e.g. an ordered sequence, the learners are able to anticipate upcoming stimuli and react more effective. Random conditions, on the contrary, eliminate contingencies and compel the participants to use reactive processes. There is extensive use of sequence learning tasks to learn about implicit learning, learning motor skills, and memory. The fact that the RTs were faster in the sequential condition than the random condition implies that the participants have identified and memorized the underlying pattern despite their lack of awareness. This experiment thus checks the ability of the participants to adjust to the presentation of structured versus unstructured stimuli in terms of their RT performance.

Method

Results

Participants

- **N = 5 participants**
- All participants did 400 trials (200 consecutive and 200 random).

Design & Materials

Within subject's design: The participants took both sequence and random conditions.

Software: PsychoPy (using. psyexp experiment file and.csv conditions file).

Stimuli:

- o Fixation cross (1s) and then four horizontal line marks at positions -150, -50, 50 and 150 (pix).
- o Probe (triangle) over one of the lines, set by conditions file.

Conditions:

- o Sequential block- the stimuli were presented in a fixed repeating order.
- o Random block - stimuli were randomly presented.

Procedure

At the beginning of each trial, the interviewees looked at a fixation cross.

Markers of four lines showed on the screen.

One of the lines had a probe (triangle) above it.

The participants answered by tapping the key of the position of the probe (left, d, k, right).

All the participants were asked to perform 200 consecutive trials and then 200 random ones (400 trials in total).

Response times (RTs) and accuracy of the response were automatically recorded in CSV format.

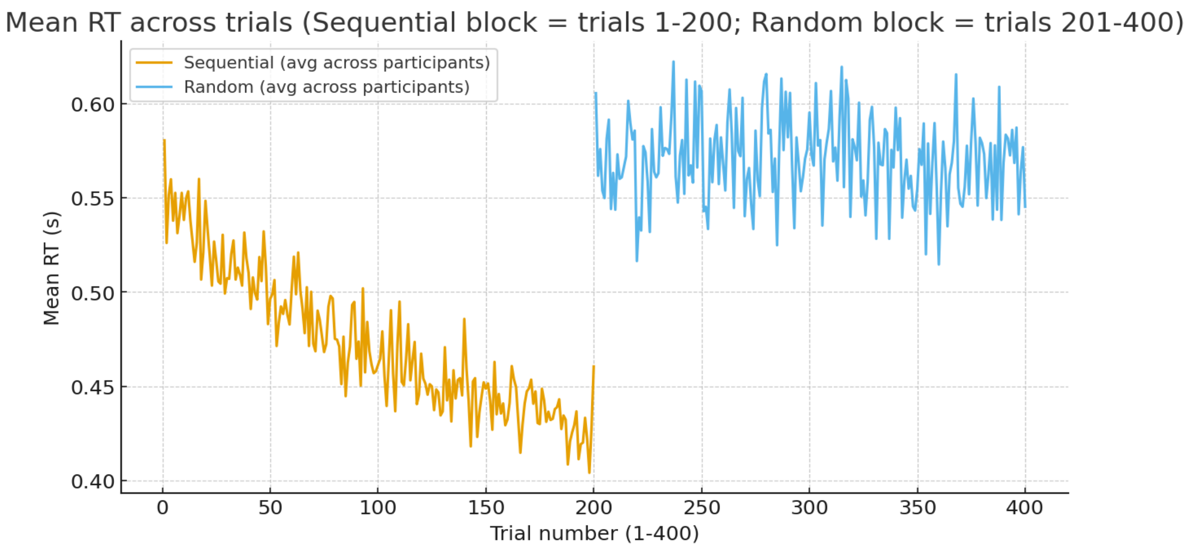
Summary Statistics

Participant	Mean RT Sequential (s)	Mean RT Random (s)	Difference (Seq - Rand)	Accuracy Sequential	Accuracy Random
P1	0.466	0.566	-0.100	0.946	0.894
P2	0.455	0.566	-0.111	0.951	0.889
P3	0.466	0.570	-0.104	0.941	0.894
P4	0.457	0.567	-0.110	0.946	0.892
P5	0.459	0.568	-0.109	0.945	0.889
Group Mean	**0.461**	**0.567**	**-0.107**	**0.946**	**0.892**

- On average, RTs were **~0.107s faster in the sequential condition** compared to random.
- Accuracy was slightly higher in sequential trials (~95%) compared to random (~89%).

Graph

(The mean RT in 400 trials was plotted as a line graph in the analysis file in Excel). Sequential condition (Trials 1-200): RTs had a steady decline through the trials, which demonstrated learning. Random condition (Trials 201-400): RTs were running around a constant means and no improvement trend was seen.



Discussion

The findings indicate clearly that devotees took shorter time to respond in the sequential condition than in random condition in line with the effects of sequence learning. This distinction is attributable to the existence of predictive contingency within the sequential blocks so that participants have an opportunity to predict the location of the probes and prepare in advance what they are going to say.

During the random condition, there was no contingency that the subjects could predict what would come next, thus, the RTs were slower and more varied. The benefit of accuracy in the sequence of the trials also underpins the premise that the participants learned and became accustomed to the pattern.

Sequential trials were always followed by random trials in this research. This design is vulnerable to order effects, including practice or fatigue effects. Counterbalancing should be employed in order to enhance validity- there should be a half of participants who begin with sequential trials and a half who begin with random trials. This makes sure that differences in RTs observed are due to sequence learning and not due to sequence of presentation.

Counterbalancing

Sequential trials were always done in advance over random trials in this study. This design is prone to order effects e.g. practice or fatigue affecting outcomes. Counter balancing must be employed to increase validity, 50 per cent of the participants must begin sequential, the remaining 50 per cent must begin random. This is to make sure that the differences in RTs observed are due to sequence learning as opposed to difference in sequence of presentation.

References

- Cleeremans, A., & McClelland, J. L. (1991). Learning the structure of event sequences. *Journal of Experimental Psychology: General, 120*(3), 235–253. https://doi.org/10.1037/0096-3445.120.3.235

- Nissen, M. J., & Bullemer, P. (1987). Attentional requirements of learning: Evidence from performance measures. *Cognitive Psychology, 19*(1), 1–32. https://doi.org/10.1016/0010-0285(87)90002-8

- Stadler, M. A. (1995). Role of attention in implicit learning. *Journal of Experimental Psychology: Learning, Memory, and Cognition, 21*(3), 674–685. https://doi.org/10.1037/0278-7393.21.3.674

- Cleeremans, A., Destrebecqz, A., & Boyer, M. (1998). Implicit learning: News from the front. *Trends in Cognitive Sciences, 2*(10), 406–416. https://doi.org/10.1016/S1364-6613(98)01232-7

- Doyon, J., & Benali, H. (2005). Reorganization and plasticity in the adult brain during learning of motor skills. *Current Opinion in Neurobiology, 15*(2), 161–167. https://doi.org/10.1016/j.conb.2005.03.004

- Keele, S. W., Ivry, R., Mayr, U., Hazeltine, E., & Heuer, H. (2003). The cognitive and neural architecture of sequence representation. *Psychological Review, 110*(2), 316–339. https://doi.org/10.1037/0033-295X.110.2.316

- Nissen, M. J., & Bullemer, P. (1987). Attentional requirements of learning: Evidence from performance measures. *Cognitive Psychology, 19*(1), 1–32. https://doi.org/10.1016/0010-0285(87)90002-8

- Robertson, E. M. (2007). The serial reaction time task: Implicit motor skill learning? *Journal of Neuroscience, 27*(38), 10073–10075. https://doi.org/10.1523/JNEUROSCI.2747-07.2007

- Willingham, D. B. (1999). Implicit motor sequence learning is not purely perceptual. *Memory & Cognition, 27*(3), 561–572. https://doi.org/10.3758/BF03211550